Deep Learning-Based Scene Text Detection and Recognition: A Comprehensive Review of CNN and Transformer Approaches

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***Abstract*—Optical Character Recognition (OCR) has evolved from rule-based character segmentation systems to modern deep learning frameworks capable of end-to-end scene text detection and recognition. Traditional OCR methods relied on handcrafted feature extraction and statistical classifiers, which performed adequately under controlled document conditions but lacked robustness in unconstrained environments. The emergence of Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and attention-based architectures significantly improved recognition accuracy by enabling automatic feature learning and sequence modeling without explicit segmentation. More recently, transformer-based models have further advanced OCR research by capturing long-range dependencies and integrating visual and textual representations within unified frameworks.**

This paper presents a comprehensive review of deep learning-based approaches for scene text detection and recognition, with particular emphasis on CNN-driven pipelines and transformer-based architectures. The study analyzes representative detection models such as EAST and CRAFT, recognition frameworks including CRNN and attention-based encoders, and recent transformer-based systems such as TrOCR. Benchmark datasets, evaluation metrics, and comparative performance trends are discussed to highlight progress and limitations. Finally, open challenges such as multilingual generalization, low-resource scenarios, domain adaptation, and computational efficiency are examined to outline future research directions in OCR and document intelligence systems.

Keywords—Optical Character Recognition, Scene Text Detection, Deep Learning, Convolutional Neural Networks, Transformers, Vision-Language Models, Document Understanding

I. Introduction

Optical Character Recognition (OCR) is a core research problem in computer vision that aims to convert textual information present in images into machine-readable format. OCR systems are widely deployed in document digitization, automated data entry, identity verification, license plate recognition, and mobile translation applications. While early OCR systems demonstrated high accuracy on scanned printed documents, performance significantly deteriorates in real-world environments where text appears under varying lighting conditions, distortions, background clutter, and arbitrary orientations [1], [2].

Traditional OCR systems followed a sequential processing pipeline consisting of image preprocessing, character segmentation, handcrafted feature extraction, and statistical classification. These approaches required precise segmentation before recognition and were highly sensitive to noise, skew, and font variability. The dependency on manually designed features limited their adaptability across languages and writing styles, particularly for handwritten text recognition [2], [3].

The introduction of deep learning marked a major transition in OCR research. Convolutional Neural Networks (CNNs) enabled automatic hierarchical feature learning directly from image data, reducing the need for manual feature engineering. Subsequently, Recurrent Neural Networks (RNNs) combined with Connectionist Temporal Classification (CTC) loss allowed recognition models to operate without explicit character segmentation, treating text as a sequence prediction problem [4], [5]. Attention-based encoder–decoder architectures further improved recognition performance by dynamically aligning visual features with output character sequences.

In parallel, scene text detection became an essential component of modern OCR systems. Unlike scanned documents, natural images contain text embedded within complex backgrounds and irregular geometries. Deep learning-based detectors such as EAST and CRAFT introduced fully convolutional frameworks capable of localizing text regions efficiently in real-time environments [6], [7]. Integration of detection and recognition models led to end-to-end text spotting systems applicable to autonomous navigation and augmented reality platforms.

More recently, transformer-based architectures have demonstrated strong performance in OCR and document understanding tasks. Vision Transformers (ViT) and transformer-based OCR systems such as TrOCR leverage self-attention mechanisms to capture long-range contextual dependencies, improving robustness in complex layouts and multilingual scenarios [8], [9]. These advancements have expanded OCR beyond isolated character recognition toward comprehensive document intelligence systems.

This paper provides a structured review of deep learning-based scene text detection and recognition methods, focusing on CNN-based pipelines and transformer-driven architectures. The remainder of this paper is organized as follows. Section II discusses traditional OCR methodologies. Section III reviews deep learning-based detection and recognition frameworks. Section IV analyzes transformer-based OCR models. Section V presents benchmark datasets and evaluation metrics. Section VI outlines research challenges and future directions. Section VII concludes the paper.

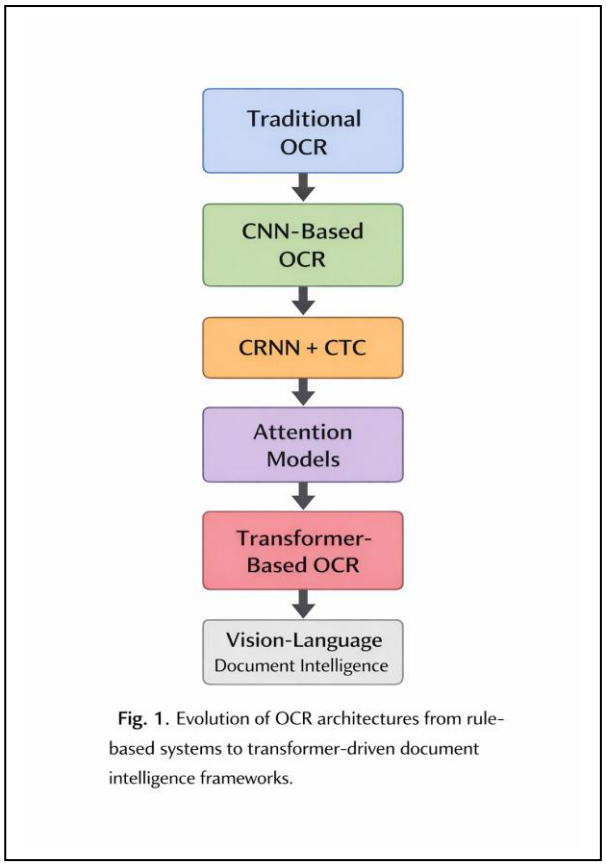

Fig. 1. Evolution of OCR architectures from rule-based systems to transformer-driven document intelligence frameworks.

II. TRADITIONAL OCR TECHNIQUES

.Before the widespread adoption of deep learning, OCR systems were primarily built upon rule-based and feature-driven pipelines. These systems followed a structured sequence of stages: preprocessing, segmentation, feature extraction, and classification. The overall objective was to isolate individual characters from document images and assign class labels using statistical pattern recognition methods.

Preprocessing played a crucial role in traditional OCR frameworks. Common techniques included grayscale conversion, binarization using thresholding algorithms such as Otsu's method, noise filtering, morphological operations, and skew correction. These steps aimed to normalize input images and enhance character visibility prior to segmentation. However, preprocessing performance was highly dependent on image quality, and errors at this stage often affected subsequent recognition accuracy [1].

Character segmentation was considered the most critical stage in classical OCR systems. Projection profiles, connected component analysis, and contour tracing were commonly used to separate characters. While segmentation worked reasonably well for printed text with uniform spacing, it became significantly more difficult in handwritten documents where characters frequently overlapped or were connected. Segmentation failures often led to cascading recognition errors, limiting system robustness [2], [3].

Feature extraction in traditional OCR relied on manually engineered descriptors. Structural features such as stroke endpoints, intersections, loops, and directional gradients were combined with statistical features including zoning, histogram of oriented gradients (HOG), projection histograms, and geometric moments. These handcrafted representations were then fed into classifiers such as k-Nearest Neighbors (k-NN), Support Vector Machines (SVM), Hidden Markov Models (HMM), and early neural networks. Although these models performed well under controlled settings, they struggled to generalize across fonts, writing styles, and languages without extensive feature redesign [2], [4].

Template matching was another widely adopted approach, particularly in early commercial OCR engines. In this method, input characters were compared against predefined templates using correlation-based similarity measures. While template-based systems achieved high accuracy for fixed-font machine-printed text, they lacked flexibility when encountering scale variations, distortions, or cursive handwriting [5].

Several early OCR engines, including initial versions of Tesseract, were largely based on segmentation-driven recognition combined with adaptive classification strategies. Although these systems achieved competitive performance for scanned documents, their dependency on handcrafted rules and segmentation accuracy limited their effectiveness in unconstrained environments such as natural scene images [5].

The primary limitations of traditional OCR approaches can be summarized as follows: (1) strong reliance on accurate character segmentation, (2) sensitivity to noise, distortions, and illumination changes, (3) limited adaptability to multilingual and cursive scripts, and (4) lack of scalability to complex scene text detection tasks. These constraints motivated the shift toward data-driven and end-to-end learning architectures, where feature extraction and sequence modeling are learned directly from data rather than manually engineered [4].

III. DEEP LEARNING-BASED SCENE TEXT DETECTION AND RECOGNITION

The limitations of traditional OCR systems led to the adoption of deep learning models capable of end-to-end

optimization. Unlike feature-engineered pipelines, deep neural networks learn hierarchical representations directly from raw image data. This section reviews major developments in CNN-based recognition models, sequence modeling approaches, attention mechanisms, and scene text detection frameworks.

A. CNN-Based Feature Learning

Convolutional Neural Networks (CNNs) introduced automatic feature extraction for OCR tasks. Instead of relying on handcrafted descriptors, CNNs learn spatial features through stacked convolutional layers and nonlinear activation functions. Early CNN-based recognition models treated character classification as a standard image classification problem, where segmented characters were directly classified into predefined categories.

Deep CNN architectures such as VGGNet and ResNet significantly improved feature robustness by increasing network depth and enabling residual learning. These models demonstrated strong performance on benchmark datasets for handwritten digit and printed character recognition tasks. However, segmentation was still required before classification, limiting their applicability to connected or cursive text [1], [2].

The need to eliminate explicit segmentation motivated the development of sequence-based models capable of recognizing entire word images directly.

B. CRNN and Connectionist Temporal Classification (CTC)

A major breakthrough in OCR research was the introduction of the Convolutional Recurrent Neural Network (CRNN), which combined CNN-based feature extraction with recurrent sequence modeling. In this architecture, convolutional layers extract visual features, which are then fed into Bidirectional Long Short-Term Memory (BiLSTM) networks to model contextual dependencies across character sequences. The output layer is trained using Connectionist Temporal Classification (CTC) loss, which allows the model to align predicted character sequences with ground truth labels without requiring explicit segmentation [3], [4].

CTC-based models treat text recognition as a sequence prediction problem. This formulation eliminates the need for character-level bounding boxes during training and enables recognition of variable-length text. CRNN models demonstrated strong performance on scene text benchmarks such as IIIT5K and ICDAR datasets. Nevertheless, CTC-based approaches assume monotonic alignment between input features and output characters, which may limit flexibility in complex layouts [3].

C. Attention-Based Encoder–Decoder Models

- To overcome the alignment limitations of CTC-based frameworks, attention-based encoder–decoder architectures were introduced. Inspired by sequence-to-sequence models in machine translation, these approaches encode visual features into latent representations and use an attention mechanism to dynamically focus on relevant regions while predicting each character.
- Attention-based OCR models improved performance in irregular text scenarios, including curved and distorted text commonly found in natural scenes. By learning soft alignment between image regions and output tokens, these models reduced dependency on monotonic assumptions. Research demonstrated improved recognition accuracy on benchmark datasets compared to pure CTC-based models, particularly for complex scene text [5].
- Despite improved flexibility, attention-based models require more computational resources and may suffer from error propagation during decoding due to autoregressive generation.

D. Deep Learning-Based Scene Text Detection

Recognition alone is insufficient in unconstrained environments; text must first be localized within images. Scene text detection aims to identify bounding regions containing text prior to recognition.

Fully convolutional detection frameworks significantly advanced scene text localization. The Efficient and Accurate Scene Text (EAST) detector introduced a geometry-based regression approach that directly predicts text region boundaries without multi-stage proposal mechanisms. EAST demonstrated real-time performance with competitive accuracy on ICDAR benchmarks [6].

Character Region Awareness for Text detection (CRAFT) further refined detection performance by modeling individual character regions and affinity relationships between adjacent characters. By predicting character-level heatmaps, CRAFT achieved improved localization accuracy, particularly for irregularly shaped text [7].

Other detection methods, including region-based and segmentation-based approaches, extended these ideas to multi-oriented and curved text detection. Deep learning-based detectors substantially outperformed classical edge-based and connected component-based detection methods.

E. End-to-End Text Spotting Systems

Modern OCR systems increasingly integrate detection and recognition into unified pipelines known as text spotting frameworks. These end-to-end architectures jointly optimize both localization and transcription tasks. Models such as Mask TextSpotter and other integrated frameworks leverage shared convolutional backbones for efficient multi-task learning.

End-to-end training reduces error propagation between detection and recognition modules and improves overall system robustness. However, these models require large annotated datasets containing both bounding boxes and transcription labels, increasing data collection complexity [8].

TABLE I Comparison of Traditional, CNN-Based, and Transformer-Based OCR Approaches

Category	Traditional OCR	CNN-RNN (CRNN)	Transformer-Based OCR

Feature Extraction	Handcrafted features	Learned via CNN	Learned via self-attention
Segmentation Requirement	Required	Not required (CTC-based)	Not required
Sequence Modeling	HMM statistical	BiLSTM	Transformer decoder
Context Modeling	Limited	Local + sequential	Global context modeling
Robustness to Distortion	Low	Moderate	High
Computational Cost	Low	Moderate	High
Multilingual Scalability	Limited	Moderate	High (with pretraining)
Layout Awareness	No	No	Yes (LayoutLM variants)

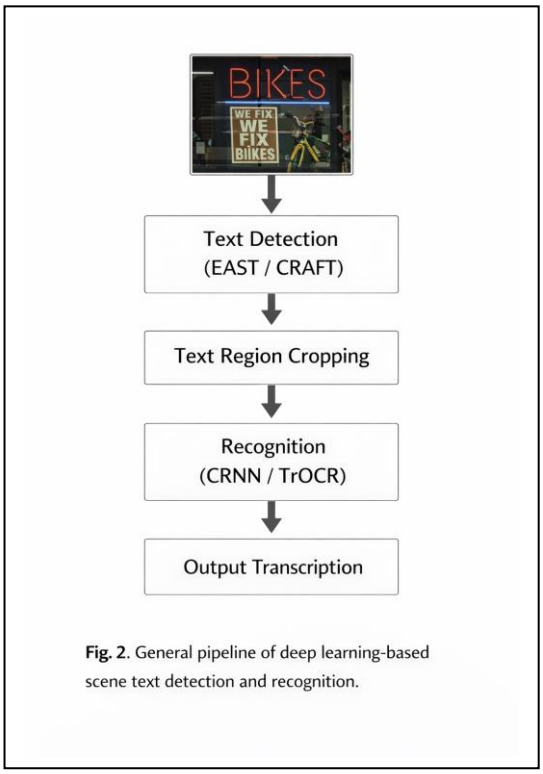

FIGURE 2 End-to-End Scene Text Spotting Pipeline

IV. TRANSFORMER-BASED OCR AND DOCUMENT UNDERSTANDING

The introduction of transformer architectures has significantly influenced computer vision and sequence modeling tasks. Unlike convolutional networks that rely on local receptive fields, transformers employ self-attention mechanisms to model global dependencies across input representations. In OCR, this capability enables improved contextual reasoning, especially for irregular layouts and long text sequences.

A. Vision Transformers in Text Recognition

Vision Transformers (ViT) adapt the transformer architecture for image processing by dividing input images into fixed-size patches, which are then linearly embedded and processed through multi-head self-attention layers. This approach allows the model to capture long-range relationships between spatial regions without convolutional inductive biases [1].

In the context of OCR, transformer-based encoders can replace CNN backbones to generate richer visual representations. By modeling global context, Vision Transformers improve robustness against distortions, perspective changes, and complex backgrounds commonly found in natural scene text [1], [2]. However, pure transformer models typically require large-scale pretraining to achieve competitive performance, which may increase computational cost.

B. Transformer-Based Text Recognition (TrOCR)

TrOCR introduced a fully transformer-based architecture for OCR that combines a vision transformer encoder with a transformer decoder for sequence generation. Unlike CRNN models that rely on recurrent layers, TrOCR leverages self-attention for both visual encoding and text decoding, enabling parallelized training and improved long-range modeling [3].

The encoder extracts patch-level features from the input image, while the decoder generates character sequences autoregressively using cross-attention. Pretraining on large synthetic text datasets significantly enhances generalization performance. Experimental results show that transformer-based OCR models achieve competitive or superior accuracy compared to CNN-RNN architectures across handwritten and printed text benchmarks [3].

One advantage of transformer-based recognition is improved flexibility in handling variable-length text and irregular alignment. Nevertheless, autoregressive decoding can introduce latency during inference, particularly for long sequences.

C. Layout-Aware and Vision-Language Models

Beyond isolated text recognition, document understanding requires modeling relationships between textual content and spatial layout. Layout-aware transformer models integrate visual features, textual embeddings, and positional information within a unified framework.LayoutLM extended the transformer architecture by incorporating 2D positional embeddings corresponding to bounding box coordinates. This design enables the model to reason about document structure while performing tasks such as form understanding and key information extraction [4]. Subsequent variants, including LayoutLMv2 and LayoutLMv3, further improved multimodal integration by jointly learning visual and textual representations [5].

These models demonstrate strong performance on structured document benchmarks such as FUNSD and RVL-CDIP. By leveraging spatial awareness, layout-aware transformers move OCR beyond simple transcription toward holistic document intelligence systems.

D. Comparison with CNN-Based Architectures

Compared to CNN-RNN pipelines, transformer-based OCR systems offer several advantages:

1. Global Context Modeling – Self-attention enables long-range dependency capture without sequential recurrence.

2. Unified Architecture – Detection, recognition, and layout reasoning can be integrated within a single framework.
3. Scalability – Large-scale pretraining enhances cross-domain adaptability.

However, transformers also introduce challenges, including higher memory requirements, longer training time, and dependence on large annotated datasets. In low-resource settings, CNN-based architectures may still offer practical advantages due to computational efficiency [2], [3].

Overall, transformer-based approaches represent a shift from isolated character recognition toward multimodal document intelligence, enabling integration with broader vision-language modeling frameworks.

V. DATASETS AND EVALUATION METRICS

Benchmark datasets play a central role in evaluating OCR systems, particularly for scene text detection and recognition tasks. Publicly available datasets enable standardized comparison of different architectures and contribute to reproducible research.

A. Scene Text Recognition Datasets

One of the most widely used benchmarks is the IIIT5K dataset, which contains cropped word images collected from the internet under varying conditions. It is commonly used to evaluate recognition accuracy under unconstrained scenarios. Similarly, the Street View Text (SVT) dataset consists of word images extracted from Google Street View, characterized by motion blur and low resolution [1].

The ICDAR Robust Reading competitions provide standardized datasets for scene text recognition and detection. Various ICDAR editions introduced increasingly challenging datasets, including multi-oriented and incidental scene text. These benchmarks have been widely adopted to evaluate both detection and recognition models [2].

For handwritten text recognition, the IAM dataset remains a standard benchmark. It contains handwritten English text samples with sentence-level annotations and is commonly used to evaluate CRNN and transformer-based recognition models [3].

B. Scene Text Detection Datasets

For text detection, ICDAR datasets are again frequently used to measure localization accuracy. Metrics such as precision, recall, and F-measure are calculated based on overlap between predicted bounding boxes and ground truth annotations.

Datasets such as MSRA-TD500 and Total-Text extend evaluation to multi-oriented and curved text detection, allowing assessment of robustness in real-world conditions. Detection models such as EAST and CRAFT are often compared using these benchmarks [2], [4].

C. Document Understanding Datasets

Beyond word-level recognition, structured document datasets are used to evaluate layout-aware models. The FUNSD dataset (Form Understanding in Noisy Scanned Documents) contains annotated form images with labeled key-value pairs, enabling evaluation of document information extraction tasks [5].

The RVL-CDIP dataset is a large-scale document classification dataset used to assess document representation learning. Layout-aware transformers such as LayoutLM are commonly evaluated on these datasets for downstream document understanding tasks [5].

D. Evaluation Metrics

OCR performance is typically measured using character-level accuracy, word-level accuracy, and edit distance metrics. Character Accuracy Rate (CAR) evaluates correct character predictions, while Word Accuracy Rate (WAR) measures exact word matches. Edit distance-based metrics such as Levenshtein distance quantify the number of insertions, deletions, and substitutions required to transform predicted text into ground truth [3].

For detection tasks, Intersection over Union (IoU), precision, recall, and F-measure are standard evaluation metrics. Detection performance is considered correct when predicted bounding boxes exceed a predefined IoU threshold with ground truth annotations [2].

Transformer-based document understanding models are often evaluated using task-specific metrics such as F1-score for key-value extraction and classification accuracy for document categorization tasks [5].

Standardized evaluation across these datasets has enabled consistent benchmarking of CNN-based and transformer-based architectures, facilitating fair comparison of model performance.

TABLE II Performance Comparison on Standard Scene Text Benchmarks

Model	**Architecture**	**Dataset**	**Word Accuracy (%)**	**Key Strength**
CRNN [9]	CNN + BiLSTM + CTC	IIIT5K	~80–85	Segmentation-free recognition
Attention OCR	CNN + Attention Decoder	SVT	~83–87	Handles irregular text
EAST [10]	Fully Conv Detection	ICDAR	F-measure ~78–80	Real-time detection
CRAFT [11]	Character-level detection	ICDAR	F-measure ~85+	Curved text localization
TrOCR [17]	ViT + Transformer Decoder	IAM	~90+	Strong contextual modeling
LayoutLM [18]	Multimodal Transformer	FUNSD	F1 ~80+	Document understanding

VI. CONCLUSION

This paper presented a comprehensive review of deep learning-based approaches for scene text detection and recognition, with particular emphasis on convolutional neural network architectures and transformer-driven models. The evolution from traditional segmentation-

based OCR systems to end-to-end learning frameworks has significantly improved recognition accuracy and robustness in unconstrained environments. CNN-based feature extraction combined with recurrent sequence modeling addressed many limitations of handcrafted pipelines, while attention mechanisms further enhanced alignment and transcription performance.

Transformer-based architectures have recently introduced global context modeling and unified multimodal representations, enabling progress toward document intelligence systems beyond isolated text recognition. Models such as TrOCR and layout-aware transformers demonstrate the potential of self-attention mechanisms for handling irregular layouts, multilingual scripts, and structured document understanding tasks.

Despite these advancements, challenges related to multilingual generalization, domain adaptation, computational efficiency, and standardized benchmarking remain open research problems. Future OCR systems are expected to integrate efficient transformer architectures with vision-language modeling techniques to support scalable and context-aware document analysis across diverse environments.

Continued research in robustness, efficiency, and multimodal integration will play a crucial role in advancing OCR technology toward practical deployment in real-world application

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